Merlin Yang

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Skills

Languages: Python, SQL, JavaScript, TypeScript
AI / LLM: Anthropic (Claude), OpenAI, Gemini APIs, **RAG**, embeddings & reranking (Voyage, Cohere), **prompt engineering**, **LLM-as-judge evaluation**
AI Agents: Claude Code **agent workflows** (**skills**, hooks, **guardrails**, plan review)
Backend & APIs: FastAPI, Express, Node.js, REST design, Pydantic
Frontend: React, Tailwind CSS
Data & Storage: PostgreSQL, Qdrant (vector database), MongoDB, SQLite, Supabase, pandas
Infra & Tooling: Docker & Compose, AWS EC2, **Git** / GitHub

Experience

FantastBioTech Inc., Full-Stack Developer · Remote 2025 – Present

- Built and shipped a full-stack AI social-media content generator end to end (Express 5 REST API, React 19 + TypeScript SPA, Supabase Postgres) for a company with no marketing function; used on their real audience, it drove a 200% increase in LinkedIn engagement in the first month.
- Designed a session-based Postgres content model and five REST endpoints, and enforced per-user data isolation with Postgres row-level security driven by forwarded JWTs rather than application code.
- Integrated OpenAI's Responses and Images APIs behind a single module with **fabrication guardrails** and defensive JSON parsing; built a two-phase image write so an interrupted upload leaves a visible failed row rather than a silent gap, and generated every post bilingually from one model call.

SiriusMindShare, Software Engineer 2022 – 2023

- Migrated and rebranded a large inherited enterprise ML codebase to the company's product line, renaming packages, modules, configs, and Oracle schemas, rebuilding **Python** wheel packages, and updating build and deploy scripts.
- Built a full-stack CSV upload feature (Angular front end + Python/Oracle backend) that upserts records on a user-defined composite key, using parameterized SQL to prevent injection.

Projects

ARC (PaperPrism) 2025 – Present

Retrieval-augmented Q&A over a researcher's 10,000-paper library; self-hosted (Mac mini, Cloudflare Tunnel, Vercel)

- Architected the **RAG** pipeline to index each paper as six representations, combine meaning and keyword search with reranking, and **verify every citation against its source chunk**; in a blind A/B over a 20-paper evaluation corpus it beat direct Claude on 11 of 15 questions. The researcher used it to write a grant proposal awarded $500,000 by the Burroughs Wellcome Fund.
- Built a retrieval **evaluation harness** on 50 real user queries, graded by a Claude Haiku 4.5 judge and scored with nDCG@10; an **index audit** caught extraction bugs that had turned 44% of chunks into fake captions and dropped every table, and old-vs-new runs on real papers verified the fix.
- Built the system with **AI agent workflows** in Claude Code: defined the architecture and **guardrails** up front, used agent **skills** and hooks, reviewed agent plans before implementation, and gated changes on side-by-side accuracy experiments.
- Diagnosed a **memory leak** that grew the service to 30 GB, traced ~90% to ML models reloaded on every upload, and fixed all six sources; it has since run for months without recurrence.

Oak Ridge National Laboratory Capstone, Team Lead 2026 – Present

Data extraction pipeline for polymer research papers

- Lead a team of three building an accuracy-critical pipeline that extracts data from polymer papers for Oak Ridge scientists; designed the architecture, built most of the backend, and coordinate work through **GitHub** issues.

jobhunt 2026 – Present

Local-first job-search pipeline with **anti-fabrication** resume tailoring

- Cut model spend by rejecting 1,041 of 1,703 discovered postings with deterministic rules before any paid LLM call (the 662 that reached a model cost $0.73 total); modeled the application lifecycle as an 11-state machine with append-only event logging and self-verifying nightly backups.

Education

Northeastern University, M.S. in Computer Science Expected Jan 2028
Northeastern University, M.P.S. in Analytics Jan 2022
University of California, Santa Cruz, B.S. in Earth Science Jan 2020